

Teaching Statement

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Over the past nine years, I've spent eleven academic terms teaching across three universities, in roles ranging from office hours TA to lead lab TA to primary instructor. I've taught or helped teach classes with enrollments ranging from 20 to 1,200, with course content varying from C++ functors to gradient descent. Throughout all my teaching experiences, my perspective has remained consistent: I always endeavor to teach the students I have, not the students I want.

The vast majority of my teaching experience is as a lab TA for large introductory classes. At Berkeley, I was an undergraduate TA for [Data 8](#), which covers Python programming and introductory statistics via simulation, for five semesters beginning in Fall 2016. As a PhD student at Michigan, I've helped teach a few classes, most notably [EECS 280](#), a CS2 course taught in C++, and [Practical Data Science](#), offered for the first time in Fall 2024, which covers real-world tools and theoretical basics for data science. In foundational courses that may be a student's first exposure to the field, a passionate, talented instructor can both open students' eyes to new pathways or career paths and help nurture the interest of students wondering if computer science is for them. One of the little things I do to make my students feel welcome is to memorize all of their names in the first week of class, so that they feel familiar and comfortable in my classroom. My students seem to notice; in the midterm course evaluations for EECS 280 in Winter 2024, one student said

"Thank you for learning everyone's names and just caring so much about the class as a whole :)"

My most significant teaching role has been as a summer instructor at UC San Diego, where I was lecturing regularly for the first time; prior to Summer 2024, I had only given guest or review lectures. Summer courses had much smaller enrollments than the regular academic year. In the first summer session, I was instructor of record for [DSC 10](#), which covers Python programming and introductory statistics using simulation. The second course I taught at UCSD was [DSC 40A](#), which covered modeling, loss function optimization, linear algebra, and probability—the mathematical foundations of learning from data. I had about 20 students in DSC 40A, and took advantage of this small enrollment to try things I ordinarily would not in a large-format class. My normal teaching approach incorporates gentle nudges to encourage students to contribute,

but I rarely directly ask someone to answer a question. Due to the course size, I tried something different in DSC 40A. I told my students that I wanted to encourage everyone to contribute to our discussion, and therefore I might directly call on them to answer questions during lectures. I made clear that they'd be able to pass on answering without any judgment whatsoever. I told my students that if anyone was hesitant about this or wanted not to be called on, they were free to reach out to me and I'd stop doing so, again without judgment. This seems to have been received well, with 100% of respondents agreeing or strongly agreeing that I "incorporated teaching methods that helped [students] learn" and several positive comments, such as:

"Often tried to engage as many people in the lecture as possible to speak up and have us work together on problems during lecture. This helped me make friends and form a study group."

"Very interactive during lecture, even if you get called on he would never discourage you if you get an answer wrong or don't know the answer. He never made you feel bad with no[t] knowing and instead helping you connect on where you don't understand."

"Encouraged group participation and told people to email him privately if they weren't okay with cold-calling."

I took advantage of the small course setting as a summer instructor to deeply engage students in problem-solving, as I had a small but attentive cohort of students in DSC 40A. In a [lecture on combinatorics](#) entirely comprised of example questions, I split the class into four groups of 3-5 students; each group worked on a set of questions and presented their answers, with supplementary explanations from me. Students responded positively to this more "flipped" lecture, noting in evaluations, for example, "We always used whiteboards to practice out the problems which were a lot more helpful than just verbal explanations" and highlighted the "Nice small class vibe." I didn't use this style in other lectures that term; I was hesitant to make too many changes to the architecture of the course without having taught it in its entirety before.



Students working in groups and presenting during the combinatorics class.

As a discussion TA for *Practical Data Science*, I did something similar when fewer students came to discussion but I knew they were engaged with the material, splitting the class into groups and asking them to present their reasoning. In the future, I aim to make my lectures as interactive as possible and will offer students low-stakes opportunities to make mistakes, ask questions, and present to their peers.

Content Development

As a summer instructor at UC San Diego, I used existing lectures and homework assignments in both courses I taught. However, prior exams for those classes had been publicly released, so my most significant content development role was in writing exams. This was the place where I had the most creative flexibility, but also the largest responsibility, since exams made up 30-50% of the course grade. I wanted to ensure my exams were clear, bug-free, and focused on the correct learning objectives. I also wanted to learn from those who had much more exam-writing experience. With that goal in mind, I ran two to three rounds of beta-testing with my TAs for every exam and sought additional feedback from instructors who had taught DSC 10 and 40A in the past. For example, in a [midterm question](#) about redundant features and multicollinearity, my first draft asked students an open-ended question about “what the result of adding a new feature will be”. I received feedback from a past instructor that the prompts were “vaguely worded” and that a more structured format would be clearer for students while assessing the same topics. I then restructured the question to include multiple-choice questions and explanation fields about, for example, the span of the design matrix or the optimal mean squared error.

Question 6

Suppose we have already fit a multiple regression hypothesis function of the form:

$$H(x) = w_0 + w_1x^{(1)} + w_2x^{(2)}$$

Now, suppose we add the feature $(x^{(1)} + x^{(2)})$ when performing multiple regression.

- a) In the space below, describe what the result of adding this new feature will be and provide rigorous mathematical justification for why this behavior might occur. Your answer must mention linear algebra concepts such as rank and linear independence in relation to the design matrix X and weight vector w^* .

- b) How many possible weight vectors w^* provide the best hypothesis function $H(x) = w_0^* + w_1^*x^{(1)} + w_2^*x^{(2)} + w_3^*(x^{(1)} + x^{(2)})$? Rigorously justify your answer.

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PID: _____

Question 6 (9.5 pts)

Suppose we have already fit a multiple regression hypothesis function of the form:

$$H(x) = w_0 + w_1x^{(1)} + w_2x^{(2)}$$

Now, suppose we add the feature $(x^{(1)} + x^{(2)})$ when performing multiple regression. Below, answer “Yes/No” to the following questions and rigorously justify why certain behavior will or will not occur. Your answer must mention linear algebra concepts such as rank and linear independence in relation to the design matrix, weight vector w^* , and hypothesis function $H(x)$.

- a) (3.5 pts) Which of the following are true about the new design matrix X_{new} with our added feature $(x^{(1)} + x^{(2)})$?
- ☐ The columns of X_{new} are linearly independent.
 - ☒ The columns of X_{new} are linearly dependent.
 - ☐ $\vec{y}$ is orthogonal to all the columns of X_{new} .
 - ☐ $\vec{y}$ is orthogonal to all the columns of the original design matrix X .
 - ☒ The columns of X_{new} have the same span as the original design matrix X .
 - ☐ $X_{new}^T X_{new}$ is a full-rank matrix.
 - ☒ $X_{new}^T X_{new}$ is not a full-rank matrix.

- b) (3 pts) Is there more than one optimal weight vector w^* that produces the lowest mean squared error hypothesis function $H(x) = w_0^* + w_1^*x^{(1)} + w_2^*x^{(2)} + w_3^*(x^{(1)} + x^{(2)})$?

☒ Yes ☐ No

Solution:

- c) (3 pts) Does the best possible mean squared error of the new hypothesis function differ from that of the previous hypothesis function?

☐ Yes ☒ No

Solution:

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Modifications to a midterm question. Old question at left, updated version at right.

This semester, I am applying and deepening my content development experience as a lead TA for EECS 280 at Michigan, implementing a new course component across the entire class: regular low-stakes quizzes. I lead the design and rollout of quizzes, given in lab sections across our enrollment of nearly 900 students. The motivation for adding quizzes arose from worsening learning outcomes over the last few years, perhaps as a consequence of increasing (but unpermitted) generative AI use by students when completing projects. Motivated by declining exam performance and a decoupling between project and exam scores, we are adding regular formative assessments to provide fast, low-stakes feedback to students outside of exams. We hope these added indicators of performance in more exam-like settings improve student performance or experience. We plan to evaluate and publish our experiences and results in CS Education venues like *SIGCSE*.

Resources & Infrastructure

Over several years of teaching, I've witnessed first-hand the "compound interest" of small efforts when teaching. Course materials may persist nearly unchanged for semesters or years, so investing the effort in building quality materials is clearly worthwhile. I've benefited from this first-hand: a basically top-to-bottom revamp of DSC 40A in the Spring 2024 quarter left me with extremely [high-quality lectures](#) I could deliver, essentially unchanged, as a summer instructor. There are more nuanced ways that small efforts compound. For example, my recommended study method for students is to take practice exams in a true exam-like setting, but often exam solutions are confusing or lack explanations; I published the exams I wrote for my Summer 2024 courses on the [DSC 10](#) and [40A Study Sites](#). I plan to continue using the open-source [Study Site](#) platform throughout my teaching career, and want to contribute to feature development like an "exam remix" option that randomly generates sample exams from a past corpus of questions. Another place where I've found that being detail-oriented pays off is in making materials portable across platforms. While it may be more work initially, building lectures or notes in LaTeX or vector graphics with TikZ enable instructors to more easily build resources in media besides slides & lectures; for example, I helped build our [FAQ page](#) for Practical Data Science, which expands on common questions and references lecture material.

Engaging Assignments

One way to better prepare students while also making assignments more relevant is to use real-world datasets. Grounding the significance of course material by using broad, representative, and intentional real-world examples makes clear the importance of

computation in everyday life and helps attract students of all backgrounds¹. For example, as a teaching assistant for *Practical Data Science*, I helped develop a new assignment (given out as a section of a longer homework) laying out the basics of retrieval-augmented generation (RAG), which identifies documents in a corpus that are most relevant to a large language model query. Production RAG uses embeddings to identify documents, but as our class covered tf-idf (term frequency–inverse document frequency), we built an assignment where students implemented tf-idf for a query like “I want to learn about machine learning”, searched over a large corpus of class syllabi for the most relevant documents, and then made an API call to a large language model asking “what classes should I take?” The most successful assignments motivate the importance of course material by drawing connections to questions students ask or tools students use. In the era of large language models in the classroom (permitted or not!), I think this assignment exemplifies that approach, and I aim to continue building assignments that accomplish this goal.

Management & Mentorship

My staff mentoring philosophy reflects my belief in small efforts that compound over time, primarily when it comes to staff onboarding. One vital resource during my summer teaching at UCSD was a staff wiki for the first course I taught. Often, these sorts of resources are ephemeral, with debugging done over Slack or pedagogical discussions taking place in weekly staff meetings; writing up instructions or recording video explanations help build a compendium of knowledge that becomes more useful over time. As a TA for *Practical Data Science*, a new class, I built a wiki with my notes on, for example, discussion worksheet compilation instructions, homework development best practices, common bugs, and teaching tips. Again, the initial investment of time may be challenging during a busy semester, but it pays off in later semesters, especially as courses grow. I’ve found that well-formalized practices are particularly important in large courses, where teaching staffs may include multiple faculty and dozens of TAs; it’s crucial to offer a consistent, high-quality instructional experience at scale, especially since the largest courses are often introductory.

I’ve mentored students in both teaching and research contexts, formally and informally. A former undergraduate mentee, Elanor, has since continued on to complete a Master’s degree in our research group and become a PhD student at Carnegie Mellon; we worked closely on the first two publications of my PhD. As the head TA and sole

¹ D. J. Ferreira, A. P. Ambrósio, T. Nogueira, M. R. D. Ullmann and T. F. N. Melo, "Students' Perceptions of Applying Real-world Problem Solving in Computer Science Education : Case Study in Interaction Design," IEEE Frontiers in Education Conference (FIE), 2018.

graduate student instructor for *Practical Data Science* at Michigan, I helped mentor our undergraduate TAs, all of whom were new to teaching discussions, via co-teaching, pedagogical discussions during office hours, and providing teaching feedback. I've served as a more informal mentor and advocate for TAs, such as Ashley, a UCSD student who led discussions for my Summer 2024 DSC 10 class. Ashley is now the sole graduate student TA, leveraging her summer discussion experience in a larger academic-year role for which I recommended her.

In large introductory classes, it's particularly important to hire a course staff that's diverse. There are obvious representational benefits to gender and racial diversity among teaching staffs: it matters for students to see themselves when they look at the instructional team². As a GSI for *Practical Data Science*, I evaluated ~40 applications and interviewed 20 TA candidates, including office hours simulations and open-ended questions; we ensured we interviewed and hired a diverse staff that was evenly gender-balanced for the Winter 2025 semester. An underrated aspect of diversity is when it comes to academic background, particularly for introductory classes like those in data science, where students may only intend on taking one or two courses. TAs who are, say, English or sociology or economics majors on a staff can contribute with novel perspectives on pedagogical strategies, applications for assignments, and ways to use computational science in the real world. Introductory courses should present this breadth of options to students; for example, as a GSI for EECS 280 at Michigan, I moderated a [Paths in Computing Panel](#) that took the place of lecture at the end of the semester. As teaching faculty, I want to offer similar opportunities for students to learn about the many options that lie before them.

The role of an instructor, especially in an introductory course, is to teach in a way that is open to the student body they serve. Each batch of new students is different, as academic disciplines such as machine learning, robotics, or data science mature, or as economic trends favor new career paths and specialties. Supporting and encouraging these students means meeting them where they are, acknowledging their backgrounds and motivations, and teaching with an eye to their circumstances. Throughout my experiences both as a teaching assistant and primary instructor, I have grown to love supporting students as they explore foundational material and new fields. As an aspiring teaching faculty member, I want to continue learning from experienced teachers, explore new topics, and share them with students.

² Fossum, T., and S. Haller. "Retaining women in CS with accessible role models." SIGCSE, 1998. and Roberts, Eric S., Marina Kassianidou, and Lilly Irani. "Encouraging women in computer science." SIGCSE, 2002.